



Department of Computer Science and Business Systems
Academic Year 2023 – 2024(Odd Semester)

10.10.2023

Degree, Semester & Branch: B.Tech., I & CSBS

Course Code & Title: GE3151 Problem Solving and Python Programming

Name of the Faculty member: Ms.K.Usharani AP/CSBS

Innovative Practice Description

- **Unit / Topic:** Unit 1 -COMPUTATIONAL THINKING AND PROBLEM SOLVING/
Illustrative Problems : Guess an integer number in a range
- **Course Outcome:** CO 1 – Develop algorithmic solutions to simple computational problems
- **Topic Learning Outcome:** TLO 3 - Students will be able to solve simple problems using program planning tools such as algorithm, pseudo code, flow chart
- **Activity Chosen:** Peer Learning
- **Justification:**
 - Peer learning fosters teamwork, cooperation, patience, and better social skills. In a cooperative peer learning environment, each student's strengths can serve to complement the group and enhance learning.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
 1. **Materials used :**
 - a. **Online Game:** https://www.mathsisfun.com/games/guess_number.html
 - b. **Reference book:** Reema Thareja, "Python Programming using problem solving approach", Second Edition, Oxford University Press, 2019
 2. **Prerequisite :** Basic knowledge on Simple Computational problem.
 3. Some volunteers are asked to play this online game in classroom before other students.
 4. All students were asked to discuss in their teams on the given problem with the help of book to build algorithm, flowchart and pseudocode.
 5. Two Interested students are asked to explain the answers to their class members.
 6. To check their knowledge on this topic, questions were asked to students.
 7. After that, the faculty consolidated the concepts once again to students and the answers for the questions were also discussed.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PO12	PSO2
CO5	2	1	1	3	3	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PO12	PSO2
	2	1	1	3	3	2	2
Justification for correlation	Apply the knowledge of engineering fundamentals like searching	Identify and analyze complex engineering problems related to searching	Design solutions for complex engineering problems with appropriate searching technique.	Function effectively as an individual and as a member in a team	Communicate effectively on complex engineering activities with engineering community.	Recognize the need and ability to engage in independent and life-long learning.	Exhibit proficiency in software development.

- **Images / Screenshot of the practice:**





- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students felt that they can able to understand and get clarity on the topic very easily by sharing their knowledge with their friends. It improves their self-learning ability. And also they felt happy while learning from game.

- ❖ ***Benefit of the practice:***

Students were actively participated in this activity.

Students interaction with their peer members and the class during the activity was good and effective.

- ❖ ***Challenges faced in implementation:***

Only few students are ready to present the topic in front of their class.

Some students are hesitated to share their knowledge in the class. And they are motivated to come out of their fear.

References:

1. <https://www.wgu.edu/blog/peer-learning2208.html#close>